Teacher's Handbook

Scientific Writing and Research Communication in Physics

B.Sc. (Honours) Physics — Semester VI

Skill Enhancement Course (SEC) — MG6SECPHY302 — 3 credits

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1. How to use this handbook

This handbook helps you plan, teach and assess the course as per the syllabus (course code MG6SECPHY302). It follows the syllabus module by module and unit by unit, keeps hours and course outcomes (COs) exactly as in the syllabus, and anchors every unit to the textbook and references the syllabus prescribes. Source names in blue are clickable links to the publisher, the open-access copy, or the official site. Anything that goes beyond the syllabus (extra sources, activity ideas, rubrics, mark splits) is marked as a *suggestion*, so you can adopt, change or drop it.

The course is a skills course. Students should leave able to read physics literature critically, find and cite reliable sources, present data honestly, and write and present physics clearly. The syllabus describes the practical component as a Scientific Writing Laboratory in which students practise, revise and peer-review their scientific communication, so the practical hours are the heart of the course, not an add-on to the lectures.

2. Course at a glance

Item	**Detail**
Programme	B.Sc. (Honours) Physics
Course	Scientific Writing and Research Communication in Physics
Course code	MG6SECPHY302
Type / Level	SEC / 300–399
Semester / Credits	VI / 3 (Lecture 2 credits, Practical 1 credit)
Contact hours	60 in total: 30 theory (Modules 1 and 2) + 30 practical (Module 3)
Prerequisites	Basic undergraduate Physics knowledge; ability to read and write in English
Teaching and learning approach	Interactive lectures, guided reading and analysis of Physics research articles, demonstrations, hands-on Scientific Writing Laboratory activities, literature and database searching, data presentation exercises, drafting and revision, peer review, poster preparation and oral presentations
Evaluation	CCA 25 marks + ESE 75 marks
ESE mode	Practical evaluation (presentation and viva): portfolio/record 20, skill assessment 25, viva 30

3. Course outcomes in the classroom

CO	Outcome	Domain	PO	What success looks like
1	Explain the nature, purpose, language and structure of scientific writing	Understand	2, 4	A student can explain why a paper is organised the way it is and can rewrite a paragraph in precise scientific style
2	Locate, organise and appropriately use authentic scientific sources in standard formats	Apply	4, 8	A student can find a primary source, judge its reliability, and cite it correctly from a managed database
3	Prepare and present scientific data accurately using appropriate tables, figures and statistical descriptions	Apply	1, 2, 4	A student can turn raw measurements into a correct table, a labelled figure with uncertainty, and a clear poster, and can report them honestly
4	Analyse research articles, reviews, posters etc. that communicate research effectively to scholarly and general audiences	Analyse	6, 9	A student can take a published item, name its elements, and explain how well each one serves its audience

CO2 and CO3 are at the Apply level and CO4 at the Analyse level. Assessment tasks should match: students should be producing and dissecting, not just recalling definitions.

4. Authentic sources

4.1 Sources prescribed by the syllabus

The syllabus names one textbook and five references.

Role	Source (click to open)	Best used for
Textbook	K. M. Klucevsek (2025), *Scientific Writing: A Guide from Data to Draft*, Duquesne University, Open Education Materials, CC BY-NC-SA	Primary text for the whole course: the route from data to a written draft. Freely accessible, and adaptable for non-commercial teaching with attribution and the same licence

Role	Source (click to open)	Best used for
Reference	A. M. Penrose and S. B. Katz, *Writing in the Sciences: Exploring Conventions of Scientific Discourse*, Parlor Press (free open-access PDF)	Why scientific writing has the conventions it has; reading papers as a genre (Module 1 and the Module 3 analysis practicals). The open edition is the third edition with some copyrighted material removed
Reference	M. Alley, *The Craft of Scientific Writing*, Springer	Style, clarity and connecting ideas at sentence and paragraph level, organising content for an audience, and using writing time well (Units 1.2, 1.5, 2.2, 2.4)
Reference	R. A. Day and B. Gastel, *How to Write and Publish a Scientific Paper*, Cambridge University Press (9th ed., 2024)	Paper structure, abstracts, titles, tables and illustrations, citation, publishing practice and ethics (Units 1.4, 2.1, 2.2, 2.4, 2.5)
Reference	J. Schimel, *Writing Science: How to Write Papers That Get Cited and Proposals That Get Funded*, Oxford University Press (catalogue record)	Story structure and argument in a paper, flow, and writing for the public (Units 1.5, 2.2, 2.4)
Reference	MIT OpenCourseWare, *Science Communication: A Practical Guide* (STS.034), CC BY-NC-SA	Open supplementary material on communicating science to varied audiences, including visualising science (Units 2.2, 2.3, 2.4)

A note on Alley. *The Craft of Scientific Writing* concentrates on writing style and process. Slide and presentation design are the subject of his companion book, *The Craft of Scientific Presentations* (Springer). That book is *not* prescribed by the syllabus; consider it an optional extra for Units 2.3 and 3.5.

4.2 Authentic sources for the practical work

The practicals ask students to work with real journals, data sources, articles, theses and popular writing. The syllabus does not list these, so treat the following as suggestions and confirm access at your institution.

- **Journals and publishers:** American Physical Society (Physical Review series); IOP Publishing; AIP Publishing; Springer Nature; Elsevier (ScienceDirect); Royal Society. Indian journals: *Pramana – Journal of Physics*, *Current Science*, and *Resonance – Journal of Science Education* (written for undergraduate students and teachers, so good for readable models).
- **Review-type journals (Unit 3.4):** *Reviews of Modern Physics*; *Reports on Progress in Physics*; *Physics Reports*.
- **Preprints, indexes and search:** arXiv; NASA ADS; INSPIRE-HEP; Crossref; DOAJ; Google Scholar.
- **Judging journal quality:** Think. Check. Submit.; Committee on Publication Ethics

(COPE).

- **Data sources:** NIST fundamental physical constants (CODATA); Particle Data Group; CERN Open Data; Gravitational Wave Open Science Center (GWOSC); NASA HEASARC; NASA open data; ESA science archives portal.
- **Theses:** Shodhganga (INFLIBNET repository of Indian theses); CERN Document Server.
- **Popular physics:** *Physics Today*; *Physics World*; *Scientific American*; APS *Physics Magazine*.
- **Tools:** Zotero and JabRef (reference managers, free and open source); Overleaf (online LaTeX); LibreOffice and Inkscape (free tools for posters and figures).
- **Access in India:** check whether your college has access through INFLIBNET or the One Nation One Subscription (ONOS) arrangement, and prefer open-access articles for class use so that every student can read the same item.
- **Licences:** the textbook and the MIT OpenCourseWare material carry Creative Commons BY-NC-SA terms: attribute, use non-commercially, and share adaptations under the same licence.

5. Planning the semester

5.1 A note on the hours

The syllabus gives 60 hours: Lecture 2 credits and Practical 1 credit. The module hours show 30 theory hours (Module 1: 10 h; Module 2: 20 h) and 30 practical hours (Module 3). A weekly pattern assumes four contact hours a week over 15 weeks, with theory and practical placed in the same week as much as possible. Each practical is placed after the theory that supports it.

5.2 Suggested sequence

Approx. weeks	Unit	Hours	CO	Type
1	1.1 Nature and purpose of scientific writing	2	1	Theory
1	1.2 Scientific language and style	2	1	Theory
2	1.3 Reading and analysing scientific articles	2	1	Theory
2	1.4 Structure of a scientific paper	2	1	Theory
3	1.5 The writing–revision cycle	2	1	Theory
3–4	2.1 Finding scientific sources, citations and references	3	2	Theory
4–5	3.1 Practical: bibliography database	4*	2	Practical
5–6	2.2 Abstract and summary, methodology and observations, case studies	4	2, 3	Theory
6–7	3.2 Practical: analyse a research article	4*	4	Practical

Approx. weeks	Unit	Hours	CO	Type
7–8	2.3 Scientific data presentations: posters, paper presentations, thesis	5	2, 3	Theory
8–9	3.3 Practical: portfolio of scientific posters	6*	4	Practical
10–11	2.4 Research articles, review articles, popular science articles, short communications	5	2, 3	Theory
11	2.5 Research ethics and academic integrity	3	3	Theory
12	3.4 Practical: analyse a topical review paper	4*	4	Practical
13	3.5 Practical: analyse a poster or presentation slides	4*	4	Practical
14	3.6 Practical: analyse a dissertation or Ph.D. thesis	4*	4	Practical
15	3.7 Practical: appreciate a popular science article or book	4*	4	Practical

*The syllabus gives 30 hours for Module 3 without dividing them between the seven practicals. The allocation shown (4, 4, 6, 4, 4, 4, 4) is a suggestion; the poster portfolio gets the most time because posters take longest to make.

Week 15 has to hold the last practical, the portfolio completion, and rehearsal for the presentation and viva. If time is short, shorten the lecture hours of Unit 2.5 or move Practical 3.7 earlier, not the practicals' feedback rounds.

6. Module 1: Foundations of Scientific Writing (10 h, CO1)

Goal: students understand what scientific writing is for, how it sounds, how papers are built, and how good writing is produced through revision.

6.1 Unit 1.1 Nature and purpose of scientific writing (2 h)

- **Key ideas:** scientific writing as communication of evidence to an audience; differences between writing for peers, for students, and for the public; accuracy, clarity, brevity, objectivity; the role of writing in the scientific process; authorship and credit.
- **Sources:** Penrose & Katz for writing as a set of community conventions; Klucevsek for the data-to-draft framing; Day & Gastel for the purpose and ethics of publication.
- **Activity:** give students two accounts of the same result, one from a research abstract and one from a news report. Ask them to list what each includes, what each leaves out, and who each is written for.
- **Evidence of learning:** a short written comparison (about 200 words).

6.2 Unit 1.2 Scientific language and style (2 h)

- **Key ideas:** precise word choice; concise sentences; active and passive voice and when each is appropriate; tense conventions; avoiding vague and inflated language; handling symbols, units and equations in running text; paragraph unity and flow.
- **Sources:** Alley for sentence-level style and connecting ideas; Klucevsek for style guidance; Day & Gastel for common errors in scientific English; Schimel for flow and energising prose.
- **Activity:** a clean-up exercise. Give a deliberately poor paragraph from a lab report (vague quantities, inconsistent units, long strings of nouns, unexplained symbols) and ask students to rewrite it in pairs.
- **Evidence of learning:** the rewritten paragraph and a list of the changes made, with reasons.
- **Teaching tip:** have students mark the subject and verb of every sentence. They quickly see where the real action is hidden.

6.3 Unit 1.3 Reading and analysing scientific articles (2 h)

- **Key ideas:** reading strategically (title, abstract, figures, conclusions, then methods); reading actively with questions; summarising in your own words; separating claims from evidence; noticing limitations.
- **Sources:** Penrose & Katz for reading a paper as a genre; Klucevsek for reading and note-taking practice.
- **Activity:** choose one short open-access physics paper (for example from arXiv or an open-access journal found through DOAJ). Students read it in three passes (skim, structure, detail) and write a one-paragraph summary, a one-sentence main claim, and two questions the paper raises.
- **Evidence of learning:** annotated paper and summary. This also prepares students for the analysis practicals in Module 3.

6.4 Unit 1.4 Structure of a scientific paper (2 h)

- **Key ideas:** title, authors and affiliations, abstract, keywords, introduction, methods (theory, experimental set-up or computation), results, discussion, conclusions, acknowledgements, references, appendices; how a physics paper adapts the standard pattern (theory and experimental papers differ); what each section must and must not do.
- **Sources:** Day & Gastel for the structure of a paper; Klucevsek for the draft-building sequence; Penrose & Katz for variation between fields.
- **Activity:** cut-up exercise. Give students the headed sections of a short paper in random order and ask them to reassemble it and justify the order. Then ask which sections they would read first and why.
- **Evidence of learning:** reassembled outline with a one-line purpose for each section.

6.5 Unit 1.5 The writing–revision cycle (2 h)

- **Key ideas:** planning, outlining, drafting, revising for structure, revising for style, proofreading; reverse outlining; giving and receiving feedback; managing time; keeping versions.
- **Sources:** Schimel for building a story and argument and for revision; Klucevsek for the process from data to draft; Alley for using writing time efficiently.
- **Activity:** students bring a short paragraph from an earlier lab report. Run two rounds of peer review using a three-question feedback sheet (What is the main point? Where did you get lost? What one change would help most?), and have the author revise after each round.
- **Evidence of learning:** three versions of the paragraph with a short note on the changes. This is the same cycle students will use throughout the lab.

7. Module 2: Scientific Literature and Data Communication (20 h, CO2 and CO3)

Goal: students can find reliable sources, cite them properly, present methods and observations truthfully, communicate in the main scientific genres, and act with integrity.

7.1 Unit 2.1 Finding scientific sources, citations and references (3 h, CO2)

- **Key ideas:** primary, secondary and tertiary sources; peer-reviewed journals, conference proceedings, preprints, books, theses, technical reports and data repositories; judging reliability (publisher, journal, authors, date, citations, methods); open access and predatory journals; search strategies (keywords, Boolean operators, citation chaining); why we cite and what needs a citation; in-text systems (numbered, author–date); reference styles common in physics (APS, AIP, IEEE; APA where a department prefers it); elements of a reference for articles, books, theses, preprints, datasets and websites; DOIs (look up with Crossref); paraphrasing and quotation (the wider ethics is in Unit 2.5).
- **Sources:** Klucevsek for finding and evaluating sources and for citing; Day & Gastel for the publication system and citation practice.
- **Tools to demonstrate:** Google Scholar, arXiv, NASA ADS, INSPIRE-HEP, Crossref, DOAJ; for journal checks, Think. Check. Submit. and COPE.
- **Activities:** a short live search on a physics topic, showing how the same query behaves in Google Scholar, arXiv, NASA ADS and a publisher database; then a repair exercise on a paragraph that uses a source badly (copied wording, missing citation, wrong style).
- **Evidence of learning:** a source-evaluation checklist applied to three sources, and a corrected paragraph with a correct reference list in one chosen style.
- **Teaching tip:** with only three hours, teach the style rules at the level Practical 3.1 needs and let the reference manager do the formatting.

7.2 Unit 2.2 Abstract and summary, presentation of methodology and observations, case studies (4 h, CO2 and CO3)

- **Key ideas:**
  - *Abstract and summary:* titles and keywords; descriptive versus informative abstracts; a plain-language summary for a general reader.
  - *Methodology:* what to report so that others can repeat the work (apparatus, procedure, calculations, software); precision without padding.
  - *Observations:* choosing between a table, a graph and text; table design (titles, units, significant figures, uncertainties); graph design (axes, labels with units, scale, error bars, fits, legends, colour-blind-safe colours); describing data in correct words (mean, standard deviation, standard error, uncertainty, regression and goodness of fit); captions that let a figure stand alone; honest presentation (no truncated or misleading axes, no selective omission of data).
  - *Case studies:* following one small piece of work, for example a student's own lab experiment or a short published paper, from raw data to abstract.
- **Sources:** Day & Gastel for abstracts, tables and illustrations; Klucevsek for the data-to-draft route and for describing data; Schimel for abstracts that tell a story; Alley for clear sentences in methods; MIT OpenCourseWare *Science Communication* for visualising science.
- **Activities (choose according to time):** (a) give students raw data from one of their own Physics lab experiments and ask them to present it as a table, a good graph and a poorly designed graph, then explain why the poor one misleads; (b) write a 150-word abstract and an 80-word plain-language summary of the same experiment and compare them; (c) peer-review one another's case studies with the feedback sheet from Unit 1.5.
- **Evidence of learning:** the three data versions with a two-sentence critique; the abstract and summary with peer comments; a one-page case study.

7.3 Unit 2.3 Scientific data presentations: posters and paper presentations, dissertation or thesis (5 h, CO2 and CO3)

- **Key ideas:**
  - *Posters:* audience and purpose; a message in the title; layout and reading order; text load; large figures; references and contact details; accessibility and colour choice.
  - *Paper presentations:* structure of a short talk (opening, message, evidence, close); slide design (one message per slide, titles that state a claim, minimal text); delivery (pace, eye contact, handling questions); presenting online.
  - *Dissertation or thesis:* purpose and audience; typical chapter structure (introduction, background, methods, results chapters, discussion, conclusion); front and back matter; how a thesis differs from a paper; the roles of supervisors and examiners; thesis repositories such as Shodhganga; long-form referencing; originality and the plagiarism check.
- **Sources:** MIT OpenCourseWare *Science Communication* for practical communication skills; Klucevsek for preparation steps and long-form planning; Penrose & Katz for

conventions of conference and extended writing; Day & Gastel for thesis writing. Optional extra (not prescribed): Alley's *The Craft of Scientific Presentations*.

- **Activities:** each student gives a two-minute talk on one finding of a paper and takes one question; the class runs a two-minute test on a sample poster; students compare the contents pages of three theses from different areas and say what each chapter title reveals about the project.
- **Evidence of learning:** a short written note on each activity, which also prepares students for Practicals 3.3, 3.5 and 3.6.

7.4 Unit 2.4 Research articles, review articles, popular science articles, short communications (5 h, CO2 and CO3)

- **Key ideas:**
  - *Research articles:* from results to story; introduction (context, gap, aim); methods and reproducibility; results versus discussion; conclusions; authorship and acknowledgements; choosing a journal, submission, peer review and responding to referee comments.
  - *Short communications:* when a letter is the right genre; limited length; what to cut and what must stay; rapid publication.
  - *Review articles:* what a review is for; types (narrative, systematic, tutorial, topical); how a review differs from a research article; using a review to enter a new topic; fair coverage and citing earlier work.
  - *Popular science articles:* audience and purpose; translating technical ideas without distorting them; analogy and its limits; narrative and human interest; headline and opening; accuracy and avoiding hype; a popular piece versus a press release; attribution.
- **Sources:** Day & Gastel for article writing, publishing and review articles; Schimel for story and argument and for writing for the public; Penrose & Katz for conventions of each genre; Klucevsek for literature synthesis and writing for general audiences; MIT OpenCourseWare *Science Communication* for popular science communication.
- **Examples to show:** review journals *Reviews of Modern Physics*, *Reports on Progress in Physics*, *Physics Reports*; popular writing in *Physics Today*, *Physics World*, APS *Physics Magazine*, *Scientific American*, and *Resonance* for pieces written for undergraduate readers.
- **Activities:** (a) take one result reported as a short communication and as a popular piece, and list what each includes, leaves out and assumes; (b) find two reviews on one topic and compare how each organises the literature; (c) rewrite the abstract written in Unit 2.2 for a school student in 150 words, then discuss what had to be dropped, simplified or kept.
- **Evidence of learning:** a comparison table for (a) and the rewritten abstract for (c).

7.5 Unit 2.5 Research ethics and academic integrity (3 h, CO3)

- **Key ideas:** honesty in data and reporting; fabrication, falsification and plagiarism (including copying one's own earlier work and duplicate publication); authorship and credit; image and data manipulation; conflicts of interest; ethics of peer review;

predatory journals; copyright and open licences (Creative Commons); responsible use of AI writing tools; correcting errors, corrections and retractions; institutional policy.

- **Sources:** Day & Gastel for ethics in publication; Klucevsek for avoiding plagiarism; COPE for publication-ethics guidance; Think. Check. Submit.
- **Activities:** case-based discussion of three short scenarios (a data point removed without comment; a paragraph copied with words changed; a gift author added); a paraphrase-versus-plagiarism exercise using a short passage from the textbook.
- **Evidence of learning:** a short reflective note, and the signed declaration at the front of the portfolio.
- **Teaching tip:** use real, well-known cases that students can read about, and focus on how the error could have been prevented.

8. Module 3: Practical (Scientific Writing Laboratory) (30 h, CO4)

Goal: students practise, revise and peer-review their scientific communication, and learn to analyse the main genres of scientific writing for expert and general readers.

The syllabus lists seven practicals and gives 30 hours for the module. The suggested allocation is below; the syllabus does not divide the hours.

Unit	Practical	Suggested hours	CO
3.1	Prepare a bibliography database	4	4
3.2	Analyse the elements of a peer-reviewed and published research article	4	4
3.3	Prepare a portfolio of scientific posters	6	4
3.4	Analyse the elements of a peer-reviewed and published topical review paper	4	4
3.5	Analyse the elements of a scientific poster or presentation slides	4	4
3.6	Analyse the elements of a dissertation or Ph.D. thesis	4	4
3.7	Appreciate the elements of a popular science article or book in physics	4	4
	Total	**30**	

Practicals 3.2 and 3.4–3.7 have the same form: students are given a real published item, read it, break it into its elements, and judge how well it communicates to its audience. Use the same analysis frame each time (Section 10.2) so that students build a transferable habit.

8.1 Unit 3.1 Practical: prepare a bibliography database (4 h)

- **Task:** build a personal reference library of at least 20 entries on a physics topic the student chooses, found using the search methods of Unit 2.1. Use a reference manager

such as Zotero or JabRef (both free and open source), generate a formatted reference list in one standard style, then switch the style. The number of entries is a suggestion.

- **Teaching points:** show how to import by DOI, how to fix wrongly imported metadata, how to tag and group references, and how to export to BibTeX or another standard format. Insist that students check every generated entry by eye; tools make mistakes.
- **Optional extension:** ask students to add a one-line reliability note beside each entry (journal, peer-review status, open access), using the checks from Unit 2.1.
- **Evidence of learning:** exported library and formatted bibliography. Portfolio item.

8.2 Unit 3.2 Practical: analyse a peer-reviewed, published research article (4 h)

- **Task:** analyse one open-access research paper in physics using the analysis frame. The student identifies each element, summarises the claim and the evidence, and comments on clarity, figures and references.
- **Choosing the paper:** a short paper of 4–8 pages from an established physics journal (see DOAJ and Section 4.2 for open-access options), in an area students can follow with a Semester VI background. Use the same paper for the whole class the first time, different papers afterwards.
- **Evidence of learning:** written analysis (about 600–800 words). Portfolio item.

8.3 Unit 3.3 Practical: prepare a portfolio of scientific posters (6 h)

- **Task:** each student prepares a portfolio of posters. A workable portfolio is at least two posters, for example one on a lab experiment or data set and one summarising a published paper in the student's own words, with correct citations.
- **Teaching points:** a poster is not a paper pasted on a board. It needs a clear message in the title, a short sequence of sections, large readable figures, minimal text, and consistent design. Provide a template in the lab and let students use free tools (for example LibreOffice Impress, Inkscape, or a LaTeX poster class on Overleaf) rather than assuming paid software. The syllabus maps this practical to CO4, but it also exercises CO3, so insist on correct tables, units and uncertainties from Unit 2.2.
- **Teacher check:** at the midpoint, run a two-minute test: a classmate who has not seen the poster must be able to state its main message after two minutes.
- **Evidence of learning:** finished posters in print or PDF. These are the central portfolio item and the usual choice for the skill-assessment presentation.

8.4 Unit 3.4 Practical: analyse a peer-reviewed, published topical review paper (4 h)

- **Task:** analyse one topical review. Identify its scope, structure, coverage of the literature, balance, figures reused with permission, and its outlook or open questions.
- **Evidence of learning:** written analysis and a one-page concept map of the review. Portfolio item.

8.5 Unit 3.5 Practical: analyse a scientific poster or presentation slides (4 h)

- **Task:** analyse a real conference poster or slide set (for example from an open conference repository or a university physics event) for message, layout, text load, figure quality, use of colour, citations and contact information.
- **Evidence of learning:** written analysis with a redesign sketch for the weakest element. Portfolio item.

8.6 Unit 3.6 Practical: analyse a dissertation or Ph.D. thesis (4 h)

- **Task:** analyse one open-access thesis (for example from Shodhganga or the CERN Document Server). A whole thesis cannot be read in 4 hours, so ask students to analyse the abstract, table of contents, introduction, one results chapter, the conclusion and the reference list.
- **Evidence of learning:** written analysis. Portfolio item.

8.7 Unit 3.7 Practical: appreciate a popular science article or book in physics (4 h)

- **Task:** read one popular article (or a chapter of a popular book) in physics and write an appreciation covering audience, hook, structure, use of analogy, accuracy, tone, and what the student learned. The syllabus says "appreciate", so the focus is on what the writing does well as well as on any weaknesses.
- **Evidence of learning:** written appreciation (about 500–700 words). Portfolio item.

9. Module 4: Teacher specific content

The syllabus leaves this module to you, so that you can choose a few of the following, or substitute your own, and fit them into practical or revision time.

- **Research ethics in more depth:** case studies that extend Unit 2.5, such as authorship disputes, image manipulation and conflicts of interest (see COPE).
- **Using AI writing tools responsibly:** what is acceptable help (spelling and grammar checks, explaining concepts) and what is not (generating text or references and submitting them as one's own). Check your institution's and the university's current policy and share it with students. Any reference suggested by a tool must be verified against the actual source.
- **LaTeX and Overleaf:** equations, figures and bibliographies.
- **Writing for the local context:** Malayalam and English science writing, school outreach, or a short piece for the college magazine.
- **A guest session** by a researcher or editor who reviews papers.

10. Practical lab management

10.1 Setting up the Scientific Writing Laboratory

- Give students equipment access: computers with internet, a printer for posters, library or database access.
- Prepare a shared folder or class workspace in which each student keeps their portfolio, with a naming convention (for example `Unit_Task_Name_Version`).
- Keep a list of open-access papers, theses and posters chosen in advance for the analysis practicals, so every student can obtain the same item.
- Give each practical a one-page instruction sheet, a deliverable, a deadline and a rubric at the start.

10.2 One analysis frame for the analysis practicals

Use these questions for each analysis (articles, reviews, posters, theses, popular pieces) so that the work is comparable and easy to mark.

Element	**Question**
Audience and purpose	Who is this for, and what is it trying to achieve?
Message	What is the one main claim or idea?
Structure	How is it organised, and does the order help the reader?
Evidence	What data, argument or example supports the message?
Visuals	Do tables, figures and layout help, and are they clearly labelled?
Language	Is the style appropriate, precise and readable?
Sources	Are the references adequate, accurate and consistently formatted?
Overall	One strength, one weakness, and one suggestion.

10.3 Peer review in the lab

Peer review is part of the course's teaching approach. Pair or group students for review rounds, give a feedback sheet, and ask reviewers to be specific and kind: one thing that works, one thing unclear, one suggestion. Mark the quality of feedback given as well as the work reviewed, so students take reviewing seriously.

11. Portfolio

The portfolio (or record) carries 20 marks in the semester-end evaluation and supplies the work for the skill assessment. A complete portfolio contains the products of the seven practical units.

Unit	Portfolio item
3.1	Bibliography database and formatted reference list
3.2	Analysis of a research article
3.3	Scientific posters
3.4	Analysis of a review paper
3.5	Analysis of a poster or slide set
3.6	Analysis of a dissertation or thesis
3.7	Appreciation of a popular science article or book

Ask students to keep earlier drafts and peer feedback in the portfolio, because the revision history shows the learning and discourages copying. A signed declaration of original work, with all sources acknowledged, goes at the front.

12. Assessment

12.1 CCA & ESE Components

Component	Marks	Options / Methods
Continuous Comprehensive Assessment (CCA)	25	Quiz, assignment, seminar, practical involvement
Semester End Evaluation (ESE)	75	Practical evaluation (presentation and viva)
ESE: portfolio / record	20	
ESE: skill assessment	25	Presentation from the portfolio
ESE: viva	30	

12.2 A suggested split of the 25 CCA marks

The syllabus names the four CCA components, you can either use all of them or choose a few among them. Here is a suggested distribution:

Component	Marks	Suggested content
Quiz	5	Short quiz after Module 1 on structure, style and ethics; another on citation styles
Assignment	7	A written task such as the abstract and plain-language summary from Unit 2.2
Seminar	7	A 5-minute presentation of a paper or a poster, followed by questions
Practical involvement	6	Attendance, participation in peer review, and timely submission of lab work

12.3 Suggested rubric for a poster

Criterion	Excellent	Adequate	Needs work
Message and title	Main point is clear in the title and at first glance	Point is present but must be searched for	Point is unclear or missing
Content accuracy	Correct physics, units and uncertainties throughout	Minor slips that do not change the meaning	Errors that affect the conclusions
Data presentation	Figures and tables clear, labelled, with units and uncertainty	Readable but with missing labels or units	Unclear, misleading or unreadable
Design and text load	Balanced layout, readable at a distance, light text	Acceptable but crowded in places	Wall of text or poor layout
Sources	Complete, accurate, consistent references	Minor inconsistencies	Missing or incorrect references
Delivery (if presented)	Clear, confident, answers questions well	Understandable but uneven	Unclear or unable to answer questions

12.4 Suggested criteria for the analysis practicals (Units 3.2, 3.4, 3.5, 3.6, 3.7)

Mark against the analysis frame in Section 10.2. A reasonable scheme: identifying the elements correctly, 30%; quality of judgement and the evidence for it, 40%; clarity of the student's own writing, 20%; correct citation of the item analysed, 10%. Insist that the analysis is in the student's own words.

12.5 The semester-end presentation and viva

- **Skill assessment (25):** the student presents one practical from the portfolio, chosen by the examiner or the student according to your department's practice. Assess content knowledge, organisation, use of visuals, delivery, and handling of questions.
- **Viva (30):** ask questions that test the whole course, not just the chosen practical. A workable pattern is six questions worth 5 marks each, drawn from different modules (suggestion).
- **Portfolio (20):** assess completeness, quality, evidence of revision, and correct citation. Check that every item is the student's own work.

Sample viva questions

- Why does a research paper have an abstract, and what should it contain?
- How would you decide whether a journal is reliable?
- What is the difference between paraphrasing and plagiarism? Give an example.

- Show a graph in your portfolio and explain what uncertainty it shows and how it was calculated.
- How does a review article differ from a research article?
- What did you change between your first and last draft, and why?
- How would you explain your poster to a school student in one minute?
- A data point does not fit your result. What may you do, and what may you not do?

13. Academic integrity

A course about scientific writing has to model good practice. This repeats in practice what Unit 2.5 teaches in theory.

- State at the start what counts as plagiarism, fabrication of data or references, and unacknowledged use of AI-generated text.
- Use a plagiarism-detection tool only as a first filter. A match is a prompt for a conversation, not proof of misconduct.
- Use different articles, theses and posters for different students where possible, so work is individual.
- Require students to supply the source link or DOI for everything they analyse.
- When open-licensed material is reused (the textbook is CC BY-NC-SA), attribute it and respect the licence terms. Check licences before reusing figures from any journal.

14. Common difficulties and how to respond

Difficulty	**Suggested response**
Students copy sentences from the source	Teach paraphrasing explicitly in Units 2.1 and 2.5 and ask for in-class summaries written without looking at the source
Weak English	Focus on short, plain sentences; use models and sentence frames; allow drafting in stages
Unreliable sources	Make the reliability checklist from Unit 2.1 a required part of every practical
Cluttered or misleading figures	Use the three-version exercise from Unit 2.2 and have students critique each other's figures
Uneven participation in group tasks	Assign roles, mark individual contributions, and keep final submissions individual
Limited database access	Use open-access journals, arXiv, DOAJ, Shodhganga and publicly available data sets
Too little time for practicals	Prepare materials in advance, set preparation as homework, and keep lab time for drafting, feedback and revision

15. Preparation checklist for the semester

- Keep the textbook handy. Download the textbook and note exact chapters against each unit.
- Get the open-access Penrose & Katz PDF.
- Choose and test the open-access articles, reviews, theses, posters and popular pieces for Units 3.2–3.7, and keep alternates in case a link fails.
- Prepare data sets for Unit 2.2 and templates for Unit 3.3.
- Install and test Zotero or JabRef on the lab computers.
- Prepare feedback sheets, rubrics and the portfolio checklist.

16. Source list

Prescribed by the syllabus

[1] K. M. Klucevsek, *Scientific Writing: A Guide from Data to Draft*, Duquesne University, Open Education Materials, 2025 (CC BY-NC-SA). https://dsc.duq.edu/oer-materials/1

[2] A. M. Penrose and S. B. Katz, *Writing in the Sciences: Exploring Conventions of Scientific Discourse*, Parlor Press (open-access edition). https://parlorpress.com/products/writing-in-the-sciences-exploring-conventions-of-scientific-discourse

[3] M. Alley, *The Craft of Scientific Writing*, Springer. https://doi.org/10.1007/978-1-4419-8288-9

[4] R. A. Day and B. Gastel, *How to Write and Publish a Scientific Paper*, Cambridge University Press, 9th ed. https://www.cambridge.org/us/universitypress/subjects/general-science/science-handbooks/how-write-and-publish-scientific-paper-8th-edition?format=PB&isbn=9781316640432

[5] J. Schimel, *Writing Science: How to Write Papers That Get Cited and Proposals That Get Funded*, Oxford University Press, 2012. Catalogue record: https://books.google.com/books?vid=ISBN9780199760244

[6] J. Durant and B. Venkataraman, *STS.034 Science Communication: A Practical Guide*, Fall 2011, MIT OpenCourseWare (CC BY-NC-SA). https://ocw.mit.edu/courses/sts-034-science-communication-a-practical-guide-fall-2011/

Optional extra (not prescribed)

[7] M. Alley, *The Craft of Scientific Presentations*, Springer. https://doi.org/10.1007/b97588

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